



Shrihari Madhusudan Dumbre

b23ee1069@iitj.ac.in

Electrical Engineering

Minor in Artificial Intelligence

Indian Institute Of Technology, Jodhpur

+91-9529076080

Codeforces

Personal Website

GitHub

linkedin.com/in/shrihari-dumbre

EDUCATION

Examination	Degree/Certificate	Institute/School	CGPA/Percentage	Year
Graduation	B.Tech. (EE)	Indian Institute of Technology, Jodhpur	7.65 (current)	2023-present
Intermediate/+2	CBSE (XII)	SNBP International School, Pune	89.2%	2023
Matriculation	ICSE (X)	Podar International School, Pune	97.1%	2021

EXPERIENCE

- Machine and Medical Vision Lab (MMVL)** *Sept. 2024 - Present*
Undergraduate Researcher IIT Jodhpur, India
 - Applied **machine learning** techniques to analyze **multimodal healthcare data** for classification and pattern discovery.
 - Explored **one-shot federated learning** to enable privacy-preserving model training across distributed medical data sources with minimal communication rounds.
 - Built scalable pipelines using **Python** and **PyTorch**; engineered features and worked with large-scale datasets.
 - Explored **dimensionality reduction** and **low-rank fusion** methods to optimize model performance and interpretability.
- Fairness and Data Efficiency Lab (FADE)** *May 2026 - Present*
Undergraduate Researcher IIT Jodhpur, India
 - Currently researching in **Class Incremental Learning (CIL)**, focusing on model forgetting and plasticity-stability trade-offs.
 - Implemented and reproduced results of **state-of-the-art papers** in CIL, benchmarking performance across standard datasets.

PROJECTS

- SLM-MoE: Small Language Model with Sparse Mixture of Experts** *Personal Project* **GitHub**
 - Built a **Sparse MoE language model** from scratch with **6 transformer layers**, **Top-2 routing** over 4 experts, **RoPE**, **RMSNorm**, and a **load-balancing loss** to prevent expert collapse; implemented **KV-cache** for faster autoregressive decoding.
 - Conducted **OOD perplexity** experiments (TinyStories → WikiText-2) and **POS affinity analysis** via forward-hook routing inspection, revealing per-expert **syntactic specialization** across layers.
- Masked Autoencoder for Self-Supervised Image Representation Learning** *Personal Project* **GitHub**
 - Implemented a **MAE** with a **Vision Transformer (ViT)** encoder trained on **CIFAR-10** using **75% patch masking**; evaluated via **linear probing (61.25%)** and **kNN classification (50.4%)**; visualized learned embeddings with **t-SNE**.
 - Deployed a **FastAPI** inference server in **Docker** exposing `/classify`, `/reconstruct`, and `/embed` endpoints, serving real-time MAE-based image classification and reconstruction.
- Low-Rank Multimodal Fusion for Gastric Histopathology Classification** *Design Credits Project under the guidance of Dr. Bikash Santra* **GitHub**
 - Developed a **multimodal PyTorch** pipeline fusing **ResNet-50** image features and **Sentence-Transformer** caption embeddings via **Low-Rank Multimodal Fusion (LMF)**, achieving **89.8% accuracy** on **PatchGastricADC22**.
 - Implemented **Attention Pooling** with a Transformer encoder for **Multiple Instance Learning** (patch → slide aggregation) and engineered the full end-to-end feature extraction and stratified data partitioning pipeline.

TECHNICAL SKILLS

- Programming:** Python, C/C++, SQL
- Frameworks and Libraries:** PyTorch, TensorFlow/Keras, NumPy, Pandas, Matplotlib, Scikit-Learn, FastAPI, HuggingFace, OpenCV, SentencePiece, Weights & Biases (W&B)
- Tools:** Git/GitHub, REST API, Google Colab, Jupyter Notebooks, Docker, Tmux
- Coursework:** Data Structures and Algorithms, Linear Algebra, Probability, Statistics and Stochastic Processes, Calculus, Pattern Recognition and Machine Learning, Computer Architecture, Natural Language Processing, ML for Communications

ACHIEVEMENTS

- Secured **All India Rank (AIR) 6002** in **JEE Advanced 2023**, among the top performers in India.
- Secured **All India Rank (AIR) 950** in **IISER Aptitude Test 2023**, among the top performers in India.
- Ranked within the **top 0.5%** out of over 1 million candidates in **JEE Mains 2023**.
- Rated **1400+ (Specialist)** on Codeforces. Best rank of 2,011 among 22,000 participants in Codeforces Round 1100.
- Achieved 99.6 percentile in the 2023 Maharashtra Common Entrance Test (MHT-CET).